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A survey of LLM-augmented knowledge graph construction and application in complex product design

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In the field of complex product design, deploying knowledge graphs (KGs) has become a promising trend due to its strength on exploiting and applying the large-scale, complex, and specialized domain knowledge. In recent years, large language models (LLMs) have also attracted much attention due to their outstanding performance in natural language understanding and generation. However, in the research of complex product design dominated by domain knowledge, few studies involve LLMs and KG at the same time. To fill this gap, we survey 42 articles published in the last four years, focusing on three key questions. The combination of LLM and KG in specific applications in complex product design is deeply discussed. The analysis reveals how these techniques facilitate data collection, design concept formation and design process optimization and proposes a technical framework combining LLM and KG for complex product design domain. In addition, we identify key challenges and propose directions for future research. As an explorative survey paper, this paper provides insightful ideas for implementing more specialized domain knowledge graph in complex product design field.

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Keywords: Knowledge Graph; Large Language Model; Design: challenges & Innovation**1. Introduction**

The design and development of complex product, such as mechanical products, aircrafts, automobiles, etc., is a knowledge-intensive task, which highly relies on the engineers' experience and the reuse of domain knowledge. Constructing a knowledge graph (KG) for domain knowledge storage, learning, and reuse has become an emerging and promising trend among design teams and companies. Myriad research studies have proved the effectiveness of KG on in-context design case retrieval, design knowledge recommendation and sharing [1], design specifications understanding [2], semantic reasoning based on textual design documents [3], and so on.

However, three issues still hinder the KG construction and applications in complex product design. Firstly, KG semantic learning requires a large dataset for training, but the knowledge of complex products is so specialized that sometimes cannot be

broadly collected. Manual labeling of training data is time-consuming for design objects of specific specialties. Secondly, to let KGs learn knowledge, a learning model needs to be coded and trained, whereas it is a hard task for a design engineer to write codes and train models. It is an urge requirement among design teams to develop a low-code way to construct and apply a domain KG. Third, the semantic understanding ability of the current domain KG model is limited, which makes it difficult to transfer to similar domains.

Fortunately, the emergence of large language models (LLMs), e.g., ChatGPT [4], could appropriately mitigate these issues. LLM could exploit pre-training techniques based on tremendous text corpora, enabling remarkable performance on comprehensive understanding and informative communications [5]. For example, MyCrunchGPT has demonstrate how ChatGPT created and generated functional code and successfully developed an interactive tool to analyze

the fluid boundary layer around a fan blade [6]. By combining the text generation capability of LLM and the structured knowledge management capability of KG, the accuracy and innovativeness of the solution can be effectively improved.

Nevertheless, there are too few literatures that use both KG and LLM technology in the current research field of complex product design, leaving many issues not been covered or fully reviewed. To fill this gap, this paper surveys around 42 articles published in the last four years. It aims to clarify the following three key research questions:

1. How can the combined use of LLMs and KGs enhance the design phase of complex engineering products?
2. How to deploy KG and LLM technologies in complex product design to organize heterogeneous data, augment complex product design capabilities, and achieve complex product design tasks with better performance?
3. What challenges and opportunities arise from the design process through LLMs and KGs?

The rest of this paper is organized as follows. The process of literature search and selection is described and illustrated in Section 2. Section 3 is devoted to addressing the first key research question, discussing collaboration ways between KGs and LLMs in complex product design. Section 4 answers the second research question, proposing an LLM-augmented-KG framework for complex product design. Section 5 delves into the third issue, discussing future prospects and challenges related to techniques for incorporating LLMs and KGs into the design process.

2. Systematic literature review process

2.1. Data source and retrieval strategy

The systematic literature selection process is depicted in Figure 1. The authors have carried out the first-round literature search using Scopus and Web of Science (WoS) databases, owing to its wide coverage of all the major peer-reviewed articles in academia. Our search strategically focuses on the concepts of KGs, LLMs, and systematic product design processes, with the aim of identifying innovative approaches to complex product design tasks, such as conceptual design, systems integration, and detailed design. The search query is formulated as follows: "Subject = ('complex product design' AND 'large language model' AND 'Knowledge graph') OR ('system product design process' AND 'LLM' AND 'knowledge engineering'); Time span: 2021-2024." This targeted search

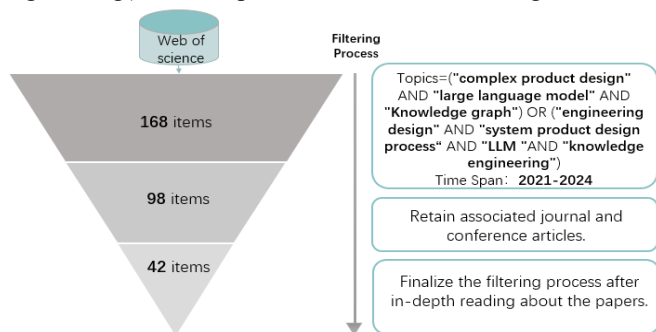


Figure 1 The systematic literature selection process

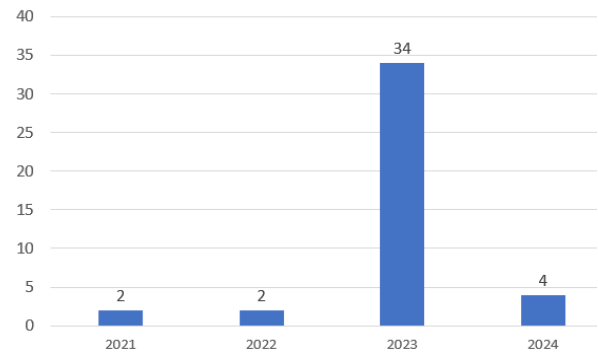


Figure 2 Number of papers per year of publication.

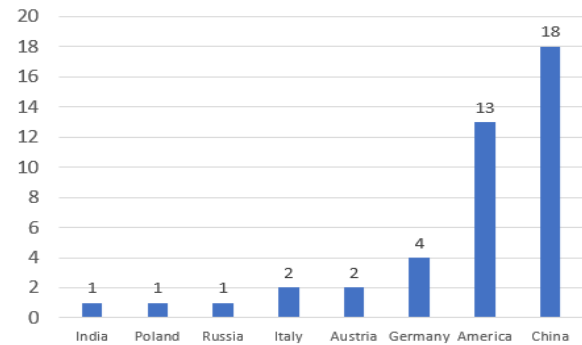


Figure 3 Country of origin of the leading institution.

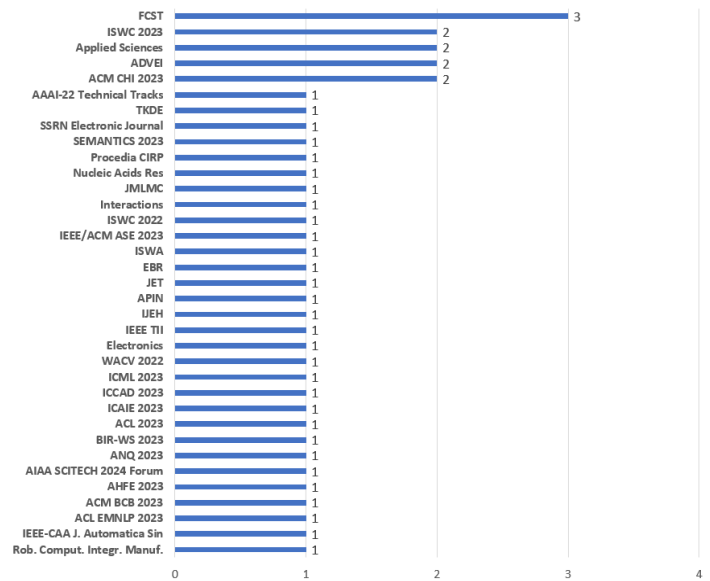


Figure 4 Publication counts by journal/conference.

was conducted on March 4, 2024 and initially yielded 168 articles. The second round of literature screening refined the database to include only articles published in academic journals and conference proceedings, narrowing the scope to 98 articles. A more detailed review was then performed by reading the content and references of the article to exclude projects that only discussed knowledge graphs and large language models separately. Ultimately, this resulted in our selection of 42 articles that provided a solid foundation for our investigation and directly contributed to our understanding of how to apply KG and LLM to enhance the design of complex products.

2.2. Statistical analysis of selected item

The 42 selected research papers were statistically analyzed. As shown in Figure 2, we can observe a significant increase in research interest in KGs and LLMs for complex product design over the last four years. Especially in 2023, the number of papers reached 34.

In terms of international collaboration, as shown in Figure 3, China leads with 18 papers, followed by the US and Germany with 13 and 4 papers, respectively. Researchers in Australia and Italy have also made significant contributions to the field, publishing two papers each. This distribution reflects that the topic of collaborating KGs and LLMs in engineering design has gained attention in countries around the globe.

Specific to the sources of publication of papers, Figure 4 shows that the Journal of Frontiers of Computer Science and Technology (FCST) has the most published sources, with three papers. The ACM and ISWC conferences respectively contributed four and three conference articles.

3. Analysis of LLM-augmented KG applications in complex product design

It is found that LLMs and KGs can facilitate lots of tasks in complex product design, including data management, decision-making support in conceptual design stage, precise question answering in embodiment design stage, and code generation in prototyping stage.

Firstly, leveraging the comprehensive knowledge mining and powerful knowledge representation capabilities of LLM and KG, data collection and organization processes can be facilitated. For instance, projects like B. Zhang [7] and Y. Wu [8] demonstrate LLM can automatically identify and extract entities, concepts, and relationships and assist to establish multi-modal dynamic knowledge graphs. In complex product design scenarios, this technique enables the extraction and integration of diverse data from sources such as sensor data, user feedback, market research, etc., resulting in a multi-modal knowledge graph relevant to the designed product. Consequently, designers not only have access to a rich knowledge base supporting innovative design schemes but also benefit from improved decision-making processes during subsequent stages.

Secondly, when it comes to decision-making support in product design workflows, combining LLM with KG empowers stronger reasoning and analytical capabilities for handling large volumes of design-related data. Notably J. Wang [9] and J. Oduro-Afriyie [10] exemplify how utilizing market insights and user sentiment stored within the knowledge graph assists LLMs in deep analysis of user needs and market trends. This addresses challenges faced by LLMs in empathizing with user emotions while aiding designers in swiftly identifying and exploring innovative concepts. Efficiency is augmented during the conceptual design phase.

Furthermore, integrating LLM with KG proves valuable for precise question answering tasks related to product appearance structure design. An example is demonstrated by the ReLMKG model [11]. By employing encoded complex questions using LLM alongside knowledge stored within KGs for answering

intricate queries based on extensive information resources. Similarly, in the detailed design task of the product, the technology can automatically answer the design specifications and historical cases integrated by the knowledge graph to provide direct design solutions and suggestions for designers, thereby enhancing the decision support in the design phase.

In the process of prototype design and testing, LLM and KG can complete the programming task of machine tools with strong logical reasoning ability and rich knowledge storage. For example: in manufacturing, KG is used to manage computer knowledge of programming and rich industrial domain knowledge to help LLMs understand and guide code generation [12]. This technology solves the problem that the generated code of LLM is difficult to adapt to the needs of complex task design due to the lack of specific engineering domain knowledge and insufficient understanding of engineering application scenarios.

Finally, the errors generated by LLM during reasoning and generation can be corrected through the factual knowledge stored in KG during the whole complex product design process. For example, X. Guan [13] and K. Węcel [14] use this method to verify the assumptions and structures of the design, which significantly improves the accuracy in the complex reasoning process and further improves the design quality.

4. A framework of LLM-augmented KG in complex product design

This section aims to answer the second research question. Figure 5 shows a proposed framework of LLM-augmented KG in complex product design, containing four layers: data layer, KG & LLM collaboration layer, augmented design capacity layer, and design task layer.

In data layer, heterogeneous product design data, including user generated data (textual data), product-sensed data (structured data), historical knowledge (rules), design cases (images or 3d models), and so on are collected. The data are multi-modal and contains semantic meanings that needed to be learned and analyzed in the following layers.

In KG & LLM collaboration layer, on the one hand, KG is the powerful tool to support design knowledge management as a knowledge base and facilitate design task reasoning as either inference chains or verification evidence. As a knowledge base, a KG could systematically organize complex product data ranging from material properties to user feedback based on their structured representation of domain-specific knowledge. The structured knowledge not only facilitates an interdisciplinary melding of insights but is dynamically scalable, ensuring up-to-the-minute relevance in a rapidly changing technological environment. By offering inference chains, reasoning models could infer following the pre-defined rules or meta knowledge, making the reasoning results explainable and thereafter convincing the engineers/designers to believe the results [15]. To further improve the confidence of the reasoning results, KGs can serve as verification device. For example, patentNet creates large-scale datasets for design patent graph recognition and retrieval, leveraging detailed patent illustrations and their associated metadata to enhance the verification capabilities of knowledge graphs [16].

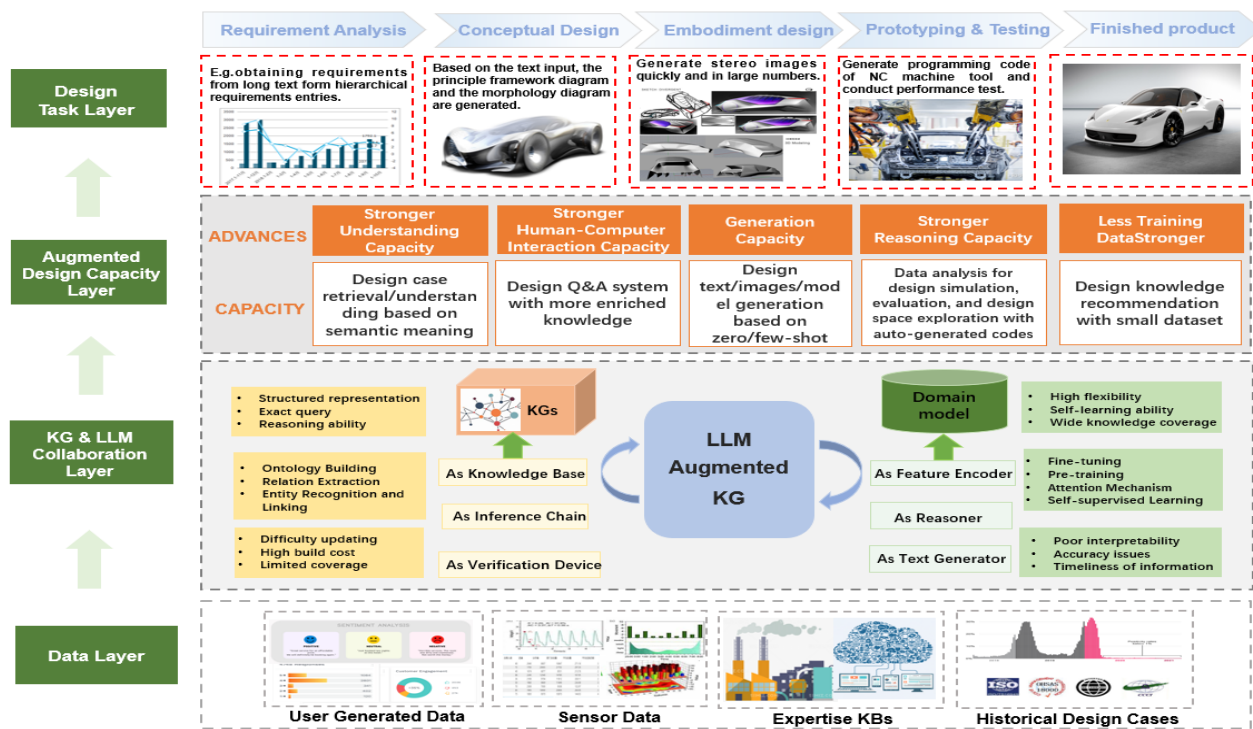


Figure 5 LLM-augmented-KG framework for complex product design.

On the other hand, LLMs are the effective technologies to handle multi-modal data and learn their meanings. Specifically, a LLM could perform as either a feature encoder, a reasoner, or a text generator. By interpreting complex queries in natural language, LLMs democratize access to vast reservoirs of information, making the design process more intuitive and accessible. They are not merely passive repositories of information but active participants in the design process by generating documents, proposing innovative solutions, and simulating design outcomes rapidly and accurately.

In the augmented design capacity layer, the collaboration of KGs and LLM further enhances various design capabilities. These include a deeper comprehension of design intricacies, intuitive and natural human-computer interaction, advanced reasoning for design decisions, generation of intricate design models and documentation, and more efficient utilization of training data. The synergy between KG and LLM enriches the understanding of explicit and tacit knowledge in product design, enabling comprehension of more complex queries and requirements. Interaction with the design system becomes more intuitive through natural language processing capabilities, facilitating a more seamless dialogue between human users and the design system. The reasoning process is supported as the system infers design implications and generates new knowledge to predict requirements and outcomes. Leveraging structured knowledge from KG, LLM's expanded generation capabilities allow it to produce text along with complex models and simulations that provide designers with a comprehensive resource to visualize and test their ideas. Finally, reliance on large amounts of training data is alleviated due to KG providing a structured knowledge base that enables LLM to make accurate predictions and recommendations with less initial input data.

The design task layer offers opportunities for building tasks in a more intelligent manner, thereby enhancing performance.

For instance, this includes integrating multi-modal data, conducting intelligent requirements analysis, facilitating advanced design activities, and providing innovative knowledge recommendation.

(1) *Integration of Multimodal Design Data:* The multimodal data including 2D images, 3D point clouds, sophisticated textual descriptions, and sensory data, is crucial in complex product design. The synergistic application of LLM and KG technologies, through machine learning models' capabilities for data fusion and semantic embedding, achieves seamless integration of diverse data modalities. LLMs are capable of interpreting textual inputs, providing contextual understanding to guide design generation, while KGs offer a structured representation of the interrelations among various data types.

(2) *Enhancing Requirement Management through Intelligent Analysis:* The analysis of user requirements and feedback greatly benefits from the natural language processing capabilities of LLMs, coupled with the structured knowledge organization of KGs. This combination permits the extraction of relevant insights from user inputs, steering the design process closer to user expectations. Implementing LLM and KG functionalities within design tools heralds a more dynamic and user-centric design methodology.

(3) *Support for Advanced Design Activities:* LLMs and KGs jointly support a spectrum of complex design activities, from early conceptual design to rapid prototyping and model-based design. By providing access to extensive repositories of design concepts and ensuring adherence to design constraints, these technologies expedite the design process while guaranteeing high-quality outcomes.

(4) *Support for Knowledge Recommendation:* The combination of semantic and relational searching enabled by LLMs and KGs offers designers a powerful tool for accessing relevant design knowledge. By understanding both the meaning behind queries and the connections between pieces of

information, designers can uncover innovative design patterns and solutions. Addressing the current gap in intuitive technology integration remains a key area for future research, aiming to develop interfaces that bring the full potential of LLM and KG technologies to the fingertips of designers in a user-friendly manner.

5. Challenges and opportunities

• *Dynamic Knowledge Update in Design.* The ever-evolving landscape of engineering data necessitates mechanisms within KGs and LLMs for real-time updates without compromising data integrity. This is crucial in complex product design, where the latest material properties, design standards, and regulatory requirements must be seamlessly integrated. Developing algorithms that can dynamically incorporate new information ensures that designers have access to the most current data, enhancing the accuracy and relevance of design decisions.

• *Cross-modal Data Integration.* Complex product design often involves diverse data types, including textual specifications, CAD drawings, and performance metrics. LLMs need to adeptly integrate these modalities, offering a unified understanding of the product. This cross-modal reasoning capability is essential for creating comprehensive design models that accurately reflect multifaceted product characteristics, facilitating a richer design process that leverages the full spectrum of available data.

• *Incorporating Domain-Specific Knowledge.* While LLMs excel in general tasks, embedding domain-specific knowledge into KGs remains a challenge. For complex product design, this involves tailoring KGs to include specialized engineering principles, material science insights, and industry-specific best practices. Enhancing LLMs and KGs with such domain-specific knowledge not only improves the relevance of generated content but also ensures that the design process benefits from deep, field-specific insights, leading to more innovative and technically sound products.

6. Conclusions

This survey summarizes recent advances in LLM-augmented KG in enhancing complex product design. A survey of 42 papers spanning a four years period found a significant increase in the number of studies in this area in 2023. Our contributions include the following four aspects:

(1) Through a detailed analysis of the years, authors and publishing institutions of the existing literature, it is clear to find that the LLM-KG collaboration is rapidly developing with increasing publication counts.

(2) Analysis of the application of LLM-augmented KG in complex product design reveals multifaceted impacts on the complex product design process, particularly through augmented data management, decision support, question answering for historical cases, and code generation. This comprehensive support system highlights the critical role of KG augmented by LLM in revolutionizing complex product design landscapes, providing a powerful framework for addressing the challenges of data complexity and improving creativity and accuracy in design decisions.

(3) An LLM-augmented-KG framework for complex product design is proposed. In this framework, KGs mainly serve as the external knowledge base for LLM, while LLMs usually work as the feature encoder, reasoner, and text generator. The integration of LLMs and KGs in complex product design offers opportunities for enhancing design processes, managing user requirements, and supporting advanced design activities through multimodal data integration and intelligent analysis.

(4) Challenges of LLM-augmented KG in engineering design are also discussed, including ensuring dynamic knowledge updates, integrating cross-modal data, and incorporating domain-specific knowledge. Addressing these challenges will unlock the full potential of these technologies in streamlining design workflows, improving decision accuracy, and fostering innovation in product development.

These insights offer a valuable view into the prospects and challenges of merging LLM with KG, guiding future research and applications. As technology advances, LLM-augmented KGs are poised to transform intelligent decision-making and creativity.

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